Law-Preserving Error Budgets for Random 3-SAT Threshold Certificates

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Abstract

The assigned manuscript separates finite negative trial certificates from the variational equality that makes their search complete. We refine the finite expectation layer without replacing its root law, literal signs, or conditional hierarchy. The two Boolean branches of an A-packet give an exact sign-count-dependent loss range, at most half the source's linear-in-count bound. Centering by the exact zero-clause values then gives asymmetric omitted-count intervals; only the B-tail is needed in the upper endpoint used to certify negativity. An ordered posterior-density argument yields depth-independent original-law L^1 message stability. Together with a range-sensitive exponent estimate, it provides an explicit all-count rationalization budget through Poisson exponential moments. We give full local proofs, exact finite fixtures, and a conditional strict-margin verification rule. The source's deletion estimates, pressure signs, structural transition and finite rational completeness remain distinct obligations. No threshold value, practical running time, or independent proof of the global computability theorem is asserted. Contribution assessment: C3/5, sharper source-specific packet and hierarchy error bounds for finite certificates, not a new threshold value or independent variational completeness proof.

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1 Model, Status and the Logical Interface

The target model has $n \geq 3$ Boolean variables and m independent proper clauses. A clause selects three distinct variables uniformly and assigns independent fair literal signs. Clauses themselves are sampled with replacement. Write $p(n, m)$ for its satisfiability probability. The source claims a limiting constant α_3 with the strict-side limits $p(n, \lfloor an \rfloor) \to 1$ below and 0 above α_3, and a single deterministic machine returning a rational within 2^{-r} on unary input 1^r [1]. Nothing in this report decides the probability at $a = \alpha_3$.

The analytic calculations instead use the source's *relaxed Poisson law*: $M \sim \mathrm{Poi}(an)$ clauses, each with three independently sampled variable positions, so within-clause repetitions are allowed. For a finite penalty $\beta > 0$, put

$$Z_n = \sum_{\sigma \in \{0,1\}^n} e^{-\beta H_n(\sigma)}, \qquad P(a, \beta) = \lim_n n^{-1} \mathbb{E} \log Z_n, \qquad w = 1 - e^{-\beta}. \tag{1}$$

The distinction between these two clause laws is not a finite-size identity. The source transfers strict-side limits by thinning and fixed density slack, followed by de-Poissonization.

1.1 Current primary context and priority

The captured source explicitly credits Gaia Carenini with priority for threshold existence. The primary ECCC record for TR26-229, published October 5, 2026, states a window $O(n^{1/2+1/k})$ and, with an Abbe–Montanari result, existence for each fixed $k \geq 3$ [2]. We checked that abstract and metadata, not its complete proof. The assigned manuscript bears a September 27 face date, while its captured bibliography explicitly includes that October 5 publication. We preserve this attribution and snapshot distinction; face dates alone do not establish priority.

The published Ding–Sly–Sun theorem concerns all sufficiently large fixed k and identifies the predicted threshold in that regime [3]. Its publisher abstract was checked; the 388-page proof was not read for this report. It is not a theorem specifically identifying the 3-SAT constant. The assigned source is analyzed as a publisher claim with authenticated cached text, not as an independently established consensus result. No value near the physics prediction, and no new rigorous numerical interval for α_3, is computed here.

1.2 Soundness is not completeness

The source's proof needs the following different assertions:

$$\ell_n \le \alpha_3, \quad \ell_n \longrightarrow \alpha_3, \quad \ell_n \text{ rational and computable,} \tag{2}$$

$$P(a,\beta) \le G(a,\beta;Q) \quad \text{for every allowed rational trial } Q, \tag{3}$$

$$P(a,\beta) = \inf_Q G(a,\beta;Q), \tag{4}$$

$$\begin{aligned} &a < \alpha_3 \Rightarrow P(a,\beta) \ge 0 \quad (\beta > 0), \\ &a > \alpha_3 \Rightarrow P(a,\beta) < 0 \quad \text{for some integer } \beta > 0. \end{aligned} \tag{5}$$

Uniform computability of each G semidecides $G < 0$ using an interval with strictly negative upper endpoint. Equations (3) and (5) then certify only $\alpha_3 \le a$, not $a > \alpha_3$. Equality need not be tested. The infimum identity (4) is what guarantees a negative witness at every rational $a > \alpha_3$.

For completeness of the logical reduction, dovetail finite descriptions, rational densities, positive integer penalties, and interval-accuracy indices; simultaneously enumerate the lower bounds. Maintain $[l,u] = [0,20]$, raising l only by a certified lower bound and lowering u only by a negative trial. Stop when $u - l \le 2^{1-r}$ and return its midpoint. Every maintained interval contains α_3. For every $\epsilon > 0$, a lower witness above $\alpha_3 - \epsilon$ eventually appears. A rational $a \in (\alpha_3, \alpha_3 + \epsilon)$ has a finite negative witness by (4)–(5), which eventually receives sufficient arithmetic precision. Thus the interval shrinks and the algorithm halts, *if all these mathematical premises hold.* The nonconstructive choice of that witness is not advice to the machine.

A single computable trial does not prove termination of this global search. Nor does termination give a usable complexity bound: the running time of a fixed total machine is itself a total computable function, by simulation, but no practical bound or measured speedup follows. This elementary computability argument is already in the source and is not a local contribution of this report.

2 Source Architecture and the Actual Trial Law

Probability and lower certificates. The source works with the capped first-unsatisfiable time $F_r = T_r \wedge 20n$ when up to r variables may be deleted. Its central uniform estimate is $\mathbb{E}(F_r^{+v} - F_r)^{6/5} \le 2^{60}$ even under an existing deletion allowance. Incident clauses split into independent obstructions for the two values of v. Residual-set conditioning replaces shorter tests and controls delay moments. Bucket resampling gives concentration; unequal-block interpolation and weight balancing give a summable merging error and

$$f_n/n \to \alpha_3, \qquad f_n/n - 2^{67} n^{-1/6} \le \alpha_3, \qquad f_n = \mathbb{E}F_0. \tag{6}$$

Finite clause-list enumeration and a Poisson survival integral make each f_n computable. The same deletion estimate supplies a positive linear density of unavoidable violations above the threshold; exact unsatisfiability alone would not imply the negative finite-temperature sign in (5).

Pressure, interpolation and reservoirs. The independent signed positions give cubic convexity and superadditivity of $\mathbb{E}\log Z_n$. A trial upper bound uses the nonnegative remainder

$$x^3 + 2y^3 - 3xy^2 = (x-y)^2(x+2y), \qquad x, y \ge 0.$$

Fresh reservoir occurrences must be conditionally independent; two occurrences in one clause still get different sites. Finite reservoirs produce nearly optimal total packet values, but initially share a nonroot Gibbs assignment and have an exponent equal to one. They are not yet allowed rational trials.

Why the equality theorem is substantial. The source preserves conditional laws for almost every *complete global path*, not merely after averaging the globals. Its exact path posterior leaves fresh site uniforms unchanged. Compensated Gaussian probes yield marked Ghirlanda–Guerra identities. Dovbysh–Sudakov, sphere support, ultrametricity and the Austin–Panchenko representation are external inputs. The local transition orders vector overlaps, chooses at most two cuts, proves marked partition independence, identifies the rescaled hidden exponents, and passes closed site-test constraints through empirical arrays and weak limits. The output horizon is fixed *before* the input horizon limit. Repeated reductions lower the first exponent or reach the root. Only afterward are nonroot shared coordinates removed using recorded kernels. Rational grids are the last approximation, not a substitute for this structural argument.

2.1 Root and site coordinates

Fix a depth $d \geq 1$ and ordered exponents $0 < t_1 \leq \cdots \leq t_d \leq 1$. Intermediate trials may have equal exponents or exponent one. Root data include a global z_0, all packet counts and literal signs, and root-site uniforms u_0^v. At level j, a general trial has a shared uniform z_j and fresh site uniforms u_j^v. Their original generators are independent. Measurable functions produce

$$f^{\pm}(z_0, \ldots, z_d; u_0^v, \ldots, u_d^v) \in [0, 1].$$

The pair is not required to sum to one. A site-only trial omits $z_1, \ldots, z_d$ but retains the shared z_0. Conditional on the root, define

$$Y_d = X, \qquad Y_{j-1} = t_j^{-1} \log \mathbb{E}[e^{t_j Y_j} \mid \mathcal{F}_{j-1}], \qquad \mathcal{T}_{\mathbf{t}} X = Y_0. \tag{7}$$

Here $\mathcal{F}_j$ records the actual coordinates through level j and the root is fixed. Rightmost transforms act first. Counts never become annealed variables inside these transforms.

For $D_A \sim \mathrm{Poi}(3a)$, each A-clause has one common real bit b, two fresh signed messages, and its own fair sign on the real bit. For $D_B \sim \mathrm{Poi}(2a)$, each B-clause has three fresh signed messages. The losses and the one-packet functional are

$$X_A = \log \sum_{b=0,1} \prod_{h=1}^{D_A} (1 - w I_h(b) f_{h1} f_{h2}), \tag{8}$$

$$X_B = \sum_{h=1}^{D_B} \log(1 - w f_{h1} f_{h2} f_{h3}), \tag{9}$$

$$G = \mathbb{E}\mathcal{T}_{\mathbf{t}} X_A - \mathbb{E}\mathcal{T}_{\mathbf{t}} X_B. \tag{10}$$

The outer expectations use the original independent root counts and signs, as well as the root/global and root-site data. In a site-only trial the total value for k packets equals kG; this is not true for arbitrary shared nonroot coordinates. A rational trial has strict rational exponents and a rational pair-valued table on finite rational rectangular grids in $(z_0, u_0, \ldots, u_d)$. These are the actual finite objects to which the source's search applies.

3 A Sign-Sensitive Packet Range

Proposition 3.1 (The two Boolean branches)**.** *Fix all root counts and signs of an A-packet. Let n_b be the number of its real literals false at b, so $n_0 + n_1 = D$. Put*

$$R_A(n_0, n_1) = \log \frac{2}{e^{-\beta n_0} + e^{-\beta n_1}} = \frac{\beta D}{2} - \log \cosh \frac{\beta(n_0 - n_1)}{2}. \tag{11}$$

For every allowed message function and every hierarchy,

$$0 \le \log 2 - \mathcal{T}X_A \le R_A(n_0, n_1) \le \frac{\beta D}{2}, \qquad 0 \le -\mathcal{T}X_B \le \beta D. \tag{12}$$

The A-range with fixed signs and the B-range are sharp within the source's allowed pair-valued message class.

Proof. Write $v_h = 1 - w f_{h1} f_{h2} \in [e^{-\beta}, 1]$. Each h belongs to exactly one of the two branches. Thus their products P_0, P_1 satisfy $P_b \ge e^{-\beta n_b}$ and $P_b \le 1$. This gives the exact lower and upper bounds for $X_A = \log(P_0 + P_1)$. Also $P_0 P_1 = \prod_h v_h \ge e^{-\beta D}$, so AM–GM gives $P_0 + P_1 \ge 2e^{-\beta D/2}$. This proves the last A-bound, or equivalently use $\cosh \ge 1$ in (11). Each B-logarithm is in $[-\beta, 0]$. Every conditional transform is monotone and preserves constants, so root-measurable bounds survive the whole hierarchy. The constant pairs $(0, 0)$ and $(1, 1)$ attain the upper and lower leaf endpoints, respectively; constant leaf values survive all transforms. In particular the second pair is admissible precisely because the source does not impose $f^+ + f^- = 1$. □

This is a strengthening of a valid source estimate, not a source correction. The source uses $\log 2 - \beta D \le X_A \le \log 2$. The sign-sensitive refinement can be much stronger than even the half-range: if all real signs agree, $R_A = \log(2/(1 + e^{-\beta D})) < \log 2$, independently of D. Both counts are zero at $D = 0$ and the range is exactly zero.

Example 3.2 (Two clauses and a constant table)**.** Take the allowed constant pair (c, c), $0 \le c \le 1$, and set $q_2 = 1 - wc^2$. For $D = 2$, opposite real signs give loss $\log 2 - X_A = -\log q_2$; equal signs give $-\log((1 + q_2^2)/2)$. At $\beta = \log 2$, $c = 1/2$, these are $\log(8/7)$ and $\log(128/113)$. The example is an exact finite packet calculation, not a negative trial certificate. The local proofs allow arbitrary $\beta > 0$; the source search enumerates positive *integer* penalties.

4 One-Sided Count Tails for a Negative Certificate

Define $L_A = \log 2 - \mathcal{T}X_A$ and $L_B = -\mathcal{T}X_B$. They are nonnegative and $G = \log 2 - \mathbb{E}L_A + \mathbb{E}L_B$. For integer cutoffs $M_A, M_B \ge 0$, retain the original, *unnormalized* probabilities in

$$G_M = \log 2 - \mathbb{E}[L_A \mathbf{1}_{\{D_A \le M_A\}}] + \mathbb{E}[L_B \mathbf{1}_{\{D_B \le M_B\}}]. \tag{13}$$

Renormalizing the retained counts would change the law and this formula. Let

$$h_\lambda(M) = \mathbb{E}_{\mathrm{Poi}(\lambda)}[D \mathbf{1}_{\{D > M\}}] = \lambda \mathbb{P}\{\mathrm{Poi}(\lambda) \ge M\}. \tag{14}$$

The equality follows by shifting $d\, e^{-\lambda} \lambda^d / d!$; it includes $M = 0$, where both sides equal λ.

Theorem 4.1 (Asymmetric omitted-count interval)**.** *For the actual trial law,*

$$G_M - \frac{\beta}{2} h_{3a}(M_A) \le G \le G_M + \beta h_{2a}(M_B). \tag{15}$$

The lower-end correction can instead use the sharper exact envelope

$$\mathbb{E}[R_A(N_0, N_1) \mathbf{1}_{\{N_0 + N_1 > M_A\}}], \qquad N_0, N_1 \text{ independent } \mathrm{Poi}(3a/2).$$

If a rational interval $[\underline{g}, \overline{g}]$ encloses G_M and a rational $b_B \ge \beta h_{2a}(M_B)$ is certified, then $\overline{g} + b_B < 0$ proves $G < 0$. No A-tail error is needed in this upper *endpoint.*

Proof. The exact difference is $G - G_M = -\mathbb{E}[L_A \mathbf{1}_{\{D_A > M_A\}}] + \mathbb{E}[L_B \mathbf{1}_{\{D_B > M_B\}}]$. Apply (12) and then (14). Independent Poisson splitting by the real sign gives the two counts in the sharper envelope. Their root law is unchanged by any of the internal transforms. The final assertion is interval inclusion followed by a strict comparison. □

This does not say that A-counts may be ignored when approximating G to arbitrary two-sided accuracy. For sound upper certification their omitted contribution is favorable; for completeness of a search using such bounds, M_A must still tend to infinity. The new errors eliminate the source's artificial $\log 2$ tail and replace its fixed-base exponential estimate by an exact size-biased Poisson tail.

For any $q \geq 1$, the classical exponential-moment bound gives

$$h_\lambda(M) \leq \lambda e^{\lambda(q-1)} q^{-M}. \tag{16}$$

For $\lambda > 0$ and $M > \lambda$, differentiation in q minimizes this expression at $q = M/\lambda$; otherwise $q = 1$ gives the trivial bound $h_\lambda(M) \leq \lambda$. If $\lambda = 0$, $h_0(M) = 0$ and no optimization is needed. For rational λ, M, the optimizing q is rational. A useful positive-series alternative, when $M + 1 > \lambda$, is

$$h_\lambda(M) \leq \lambda e^{-\lambda} \frac{\lambda^M}{M!} \frac{1}{1 - \lambda/(M+1)}. \tag{17}$$

It follows by bounding the ratios of successive Poisson terms from index M onward. Exact finite sums also give $h_\lambda(M) = \lambda(1 - e^{-\lambda} \sum_{j=0}^{M-1} \lambda^j / j!)$ for $M \geq 1$. Directed rational enclosures must account for subtractive cancellation; (17) avoids it. These Poisson facts are classical. The increment here is their sign-aware use in this particular A-minus-B certificate.

5 Original-Law Stability Without a Depth Factor

The following estimate uses the ordered-exponent density calculation already present in the source sensitivity proof. Its application here is to *leaf error under the original law*, not to an independently resampled posterior or to a change in the kernels.

Theorem 5.1 (Depth-independent leaf stability)**.** *Fix the conditional kernels and root value in* (7)*. Suppose X, X' take values in a common interval $[l, u]$ of width $R = u - l$ and $0 < t_1 \leq \cdots \leq t_d \leq t_* \leq 1$. Then*

$$|\mathcal{T}_{\mathbf{t}} X' - \mathcal{T}_{\mathbf{t}} X| \leq \min\{\|X' - X\|_\infty,\ e^{t_* R} \mathbb{E}|X' - X|\}. \tag{18}$$

The expectation is under the unchanged original conditional path law. All bounds hold for almost every root value.

Proof. Monotonicity and translation equivariance give the supremum bound. For the mean bound interpolate $X_s = X + s(X' - X)$, $0 \leq s \leq 1$, and let Y_j^s be its effective values. Every $Y_j^s \in [l, u]$. The derivative of a one-step logarithmic average in its argument is its normalized exponential average. The chain rule therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}s} \mathcal{T}_{\mathbf{t}} X_s = \mathbb{E}[W_d^s (X' - X)], \qquad W_b^s = \exp\left\{ \sum_{j=1}^{b} t_j (Y_j^s - Y_{j-1}^s) \right\}. \tag{19}$$

Each successive factor has conditional mean one under its original kernel; W_b^s is the density of the actual successive tilted history law, and $\mathbb{E} W_b^s = 1$. This is not a product of independent effective

values. Summation by parts gives

$$\log W_b^s = t_b Y_b^s - t_1 Y_0^s - \sum_{j=1}^{b-1} (t_{j+1} - t_j) Y_j^s. \tag{20}$$

The coefficients after the first are nonpositive and their absolute values sum to t_b. Consequently $|\log W_b^s| \le t_b R$. In particular $W_d^s \le e^{t_* R}$, independently of d. Integrating (19) proves (18). Boundedness justifies conditional differentiation and Fubini. The same proof works for general standard Borel kernels; it requires no finite grid. □

Using an e^R factor at every level would produce an artificial e^{dR} cost. Equation (20) removes that accumulation for fixed ordered exponents. It does not remove the number of entries in a finite tree evaluation, nor does it prove the source's kernel-collapse or exchangeability theorem. It also does not allow changing the original conditional laws while using the old mean error.

5.1 Changing the exponents

Here a range-sensitive refinement of the source's own sensitivity lemma is useful for a complete rationalization budget.

Proposition 5.2 (Range-sensitive exponent cost)**.** *Under fixed kernels, let $X \in [l, u]$, $R = u - l$, and let $\mathbf{t}, \mathbf{t}'$ be two nondecreasing vectors in $(0, t_*]^d$. Then*

$$|\mathcal{T}_{\mathbf{t}} X - \mathcal{T}_{\mathbf{t}'} X| \le \frac{R^2}{2} e^{2t_* R} \|\mathbf{t} - \mathbf{t}'\|_\infty. \tag{21}$$

For a single operation, including its continuous value at zero,

$$0 \le T_t X - \mathbb{E}X \le tR^2/8, \qquad t \ge 0. \tag{22}$$

Proof. Translation equivariance lets us assume $0 \le X \le R$. For a fixed parameter vector, set $D_j = Y_{j-1} - \mathbb{E}[Y_j \mid \mathcal{F}_{j-1}] \ge 0$ by Jensen. The original-law variance budget sharpens to

$$\begin{aligned} \sum_j \mathbb{E} \operatorname{Var}(Y_j \mid \mathcal{F}_{j-1}) &\le \mathbb{E}X^2 - Y_0^2 + 2R(Y_0 - \mathbb{E}X) \\ &= \mathbb{E}(X - R)^2 - (Y_0 - R)^2 \le R^2. \end{aligned} \tag{23}$$

Indeed conditional variance is no larger than the conditional mean of $(Y_j - Y_{j-1})^2$; expand that square, use $Y_{j-1} \le R$, and telescope.

If $\psi(t) = \log \mathbb{E}e^{tV}$ for a variable with range width R, then $\psi''(t) = \operatorname{Var}_t(V)$ and

$$\frac{\mathrm{d}}{\mathrm{d}t} T_t V = t^{-2} \int_0^t v \operatorname{Var}_v(V) \, \mathrm{d}v.$$

For $0 \le v \le t_*$ the tilted density is at most $e^{t_* R}$; using the untilted mean as a competitor gives $\operatorname{Var}_v(V) \le e^{t_* R} \operatorname{Var}(V)$. Differentiating the jth exponent and propagating outward therefore gives a nonnegative partial derivative bounded by $\frac{1}{2} e^{2t_* R} \mathbb{E} \operatorname{Var}(Y_j \mid \mathcal{F}_{j-1})$: one exponential factor comes from the local variance comparison, the other from W_{j-1} in (20). Sum, apply (23), and integrate along the straight segment of ordered vectors. This proves (21).

For (22), every tilted law is still supported on $[l, u]$, so its variance is at most $R^2/4$. This bound follows from $\mathbb{E}(V - l)(u - V) \ge 0$ and $(\mathbb{E}V - l)(u - \mathbb{E}V) \le R^2/4$. Twice integrating $\psi''(v) \le R^2/4$ gives $\psi(t) - t\mathbb{E}X \le t^2 R^2/8$, while Jensen gives the lower bound. This is the classical Hoeffding range estimate. □

The source uses $K(C) = \frac{5}{2}C^2e^{4C}$ for $|X| \le C$. Centering first gives $C = R/2$; the variance telescope above further reduces the resulting factor $5/8$ to $1/2$. For A-packets we additionally have $R \le \beta D/2$; for B-packets $R \le \beta D$. These are refinements of the source calculation, not newly attributed general concentration laws.

6 An Explicit All-Count Rationalization Bound

Let $f, \widetilde{f}$ be two pair-valued message maps on the *same* root/site generator space, with the same finite depth and unchanged kernels. In particular a shared root remains shared. Write

$$\xi = \frac{1}{2}\mathbb{E}_{\text{base}}(|f^+ - \widetilde{f}^+| + |f^- - \widetilde{f}^-|), \qquad c_\beta = e^\beta - 1. \tag{24}$$

The base expectation includes the root global and one site's entire original path, but not any fresh packet reweighting. It is also the expected error in one message chosen by a fresh fair literal sign. A pointwise componentwise error is denoted by $\eta = \max_\pm \|f^\pm - \widetilde{f}^\pm\|_\infty$.

Define the Poisson exponential moments

$$F_\lambda(c) = \lambda e^c \exp\{\lambda(e^c - 1)\} = \mathbb{E}_{\text{Poi}(\lambda)} D e^{cD}, \tag{25}$$

$$H_\lambda(c) = (\lambda e^c + \lambda^2 e^{2c}) \exp\{\lambda(e^c - 1)\} = \mathbb{E}_{\text{Poi}(\lambda)} D^2 e^{cD}. \tag{26}$$

They follow by one and two derivatives of the Poisson generating function, or directly by shifting its positive series.

Theorem 6.1 (Finite-hierarchy, all-count error)**.** *Assume the two maps are $[0,1]^2$-valued, with the same fixed conditional kernels, independent original root counts and signs, and nondecreasing exponent vectors bounded by $t_* \le 1$. Let $\zeta = \|\mathbf{t} - \widetilde{\mathbf{t}}\|_\infty$. For their fully averaged one-packet values,*

$$|G(f, \mathbf{t}) - G(\widetilde{f}, \widetilde{\mathbf{t}})| \le C_{\text{msg}}\xi + C_{\text{exp}}\zeta, \tag{27}$$

where the finite constants are

$$C_{\text{msg}} = c_\beta\{2F_{3a}(t_*\beta/2) + 3F_{2a}(t_*\beta)\}, \tag{28}$$

$$C_{\text{exp}} = \frac{\beta^2}{8}H_{3a}(t_*\beta) + \frac{\beta^2}{2}H_{2a}(2t_*\beta). \tag{29}$$

Neither constant depends on the depth. For a certified uniform message error, $C_{\text{msg}}\xi$ may instead be replaced by

$$12ac_\beta\eta. \tag{30}$$

One may take the smaller of the two message bounds when both premises are available.

Proof. For $v, \widetilde{v} \in [0,1]^k$, $k = 2, 3$,

$$|\log(1 - w\prod v_i) - \log(1 - w\prod \widetilde{v}_i)| \le c_\beta \sum_i |v_i - \widetilde{v}_i|.$$

This follows by the product telescoping inequality and the derivative bound $w/(1-w) = c_\beta$. Log-sum-exp is nonexpansive for the maximum error in its two branch log-products. Thus the A leaf error is bounded by c_β times the sum over its $2D_A$ message occurrences; the B leaf error is bounded by the analogous $3D_B$ sum.

Fix root counts and signs before applying Theorem 5.1. Both A losses lie in the same sign-dependent interval (11); it suffices to use $R = \beta D_A/2$. Both B losses lie in $[-\beta D_B, 0]$. Integrate the resulting original-law mean errors over root and sign data. Every reservoir sign is fresh and fair and is independent of its generators. Linearity of expectation gives the sum of the individual errors even though their root global is common. Counts are independent of these maps. Hence

$$|\Delta G_{\mathrm{msg}}| \le c_\beta \xi \{\mathbb{E}[2D_A e^{t_* \beta D_A/2}] + \mathbb{E}[3D_B e^{t_* \beta D_B}]\},$$

which is (28). With a uniform componentwise error, use the supremum part of (18) instead and average $2D_A + 3D_B$, obtaining $2(3a) + 3(2a) = 12a$.

Next hold the message map fixed and change the exponents. Proposition 5.2 gives the A cost $\zeta\beta^2 D_A^2 e^{t_* \beta D_A}/8$ and the B cost $\zeta\beta^2 D_B^2 e^{2t_* \beta D_B}/2$. Their Poisson expectations give (29). Split a simultaneous change into these two operations. The range bounds hold for both maps, so the same constants apply. All integrations are justified by the displayed finite positive Poisson exponential moments. □

For a measurable site-only map, dyadic conditional expectations and range-preserving rational quantization give L^1 convergence to rational tables. Theorem 6.1 supplies an explicit modulus *in the certified L^1 error* for the trial value, without first truncating counts or paying once per hierarchy level. It does not supply an algorithm to compute an L^1 approximation rate for an arbitrary unknown measurable kernel. Similarly, it does not construct the source's conditional type laws or convert a shared-nonroot trial to a site-only trial. Those are earlier, genuinely different steps in completeness.

The constants can be very large, especially at large β. For practice one can use the all-count bound, a cutoff-dependent bound, or the uniform estimate (30), whichever is certified and smaller. Independence of depth is an error statement, not a claim of polynomial evaluation cost.

6.1 A law-preserving strict-margin rule

Corollary 6.2. *Let Q be a valid rational site-only trial to which the source's upper-bound theorem applies. Evaluate an approximation $\widetilde{Q}$ on the same generator space with certified message and exponent errors ξ, ζ. Let $[\underline{g}, \overline{g}]$ enclose the truncated value (13) of $\widetilde{Q}$ and let b_B upper-bound its B-tail as in Theorem 4.1. If*

$$\overline{g} + b_B + C_{\mathrm{msg}}\xi + C_{\mathrm{exp}}\zeta < 0, \tag{31}$$

then $G(Q) < 0$. Conditional on the source soundness and below-threshold pressure sign, this certifies $\alpha_3 \le a$.

Proof. Theorem 4.1 bounds $G(\widetilde{Q})$ from above. Theorem 6.1 bounds its difference from $G(Q)$. The strict inequality proves the claim. An approximate table need not itself be used as an upper-bound trial when only this error comparison is invoked; the original Q must satisfy the source theorem's full premises. □

For exact interval evaluation of Q, take $\xi = \zeta = 0$. No theorem above makes a zero-margin witness decidable. For example, no finite precision index is guaranteed to recognize $G = 0$ by strict negative upper endpoints.

7 Finite Expectations and Original Ram Comparisons

7.1 Exact tables before surrogate quadrature

For fixed counts and signs, a rational trial has finitely many rectangles. Conditional integration at a node is a finite sum with its *actual* rational rectangle masses. The full expression uses positive sums, exponentials and logarithms. Clause factors are at least $e^{-\beta}$; every divided logarithm has a positive denominator parameter. Certified rational arithmetic therefore encloses the expression, however costly its expansion may be.

The original local-moment proof [4] derives its finite Taylor identity by integrating against the specified kernel and bounding an absolute-moment remainder. It does not replace that kernel by a convenient probability distribution. For illustration, at one node let $X = c + V$, $|V| \le R/2$, and let $m_j = \mathbb{E}_\mu V^j$ under the *actual conditional law*. Then

$$I(t) = \mathbb{E}_\mu e^{tV}, \qquad A_N(t) = \sum_{j=0}^{N} t^j m_j / j!, \tag{32}$$

$$|I(t) - A_N(t)| \le e^{tR/2} \frac{(tR/2)^{N+1}}{(N+1)!} =: \epsilon_N(t). \tag{33}$$

This is the original finite-remainder argument specialized to the exponential, not a new general moment theorem. If moment intervals have radii ϵ_j, add $\sum_{j=0}^{N} t^j \epsilon_j / j!$ to the right side. If a chosen approximation $\widehat{I}$ has total error $\epsilon < e^{-tR/2}$, its positivity is certified, and

$$\left| t^{-1} \log I - t^{-1} \log \widehat{I} \right| \le \frac{\epsilon}{t(e^{-tR/2} - \epsilon)}. \tag{34}$$

The denominator follows from the lower bound on both arguments and the derivative of log. For $N \ge 1$ the Taylor part of this error behaves as $O(t^N)$ as $t \downarrow 0$; a fixed arithmetic error independent of t does not have that property. Every downstream node needs its own valid conditional error; one may not reuse unconditional moments after a nonlinear tilt.

The original TR250 discussion [5] defines moments as functionals on the chosen physical space and distinguishes unisolvence there from a bound on an infinite-dimensional space. Its positive-weight polynomial construction likewise does not justify changing a probability law. In the present problem, finite rational step tables are already an exact description of an admissible trial. Polynomial fitting is optional and requires the explicit remainder and positivity conditions above. No PDE coordinate-transfer identity proves (4).

Example 7.1 (The same site marginal is not the same trial)**.** Take a root bit Z uniform on $\{0, 1\}$ and let every site's message pair equal (Z, Z); later site variables are unused. For a B-packet with one clause, the mean loss is $-\beta/2$. If the common root is replaced by three independent bits with the same one-site marginal, the mean loss becomes $-\beta/8$. Indeed the product of the three messages equals one with probability $1/2$ in the first law and $1/8$ in the second. All individual message marginals agree, yet the error is $3\beta/8$. Theorems 5.1 and 6.1 retain the common generator space precisely to exclude this change. The source's pathwise continuation records address a more demanding dependence-removal task; they cannot be replaced by agreement of one-site marginals.

Likewise a variable equally likely to be zero or one and the constant $1/2$ have the same first moment, but their T_1 values are $\log((1 + e)/2)$ and $1/2$, respectively, with a strict gap. A finite moment match without a controlled remainder is not an equality of entropic averages.

7.2 A reduced trial that keeps the correct count law

For the constant pair (c, c), every hierarchy is irrelevant, but the root signs and counts are not. Independent Poisson splitting gives $N_0, N_1 \sim \mathrm{Poi}(3a/2)$ and

$$G_c = \mathbb{E}\log(q_2^{N_0} + q_2^{N_1}) - 2a\log q_3, \quad q_2 = 1 - wc^2, \quad q_3 = 1 - wc^3. \tag{35}$$

This is an actual two-dimensional positive Poisson expectation, not a Gaussian approximation to it. It provides a small test interface for (15). In particular $G_0 = \log 2$, while

$$G_1 = \log 2 + \frac{a\beta}{2} + \mathbb{E}\log\cosh\frac{\beta(N_0 - N_1)}{2} > 0. \tag{36}$$

Thus neither extreme constant table gives a negative witness. These sanity checks do not evaluate the infimum over all trials. We have not run the global enumeration or claimed a new upper bound from the intermediate table $c = 1/2$.

8 Comparison, Costs and Remaining Research

The local results preserve the source's real-bit sum, independent literal signs, fresh occurrences, common root, positive clause factors and ordered conditional transforms. They improve three finite error mechanisms:

1. The A-packet range is sign-aware and at most $\beta D/2$, instead of the source's βD interval width.

2. The exact centering produces an asymmetric count error. For strict upper certification only the unfavorable B-tail is charged. Exact size-biased or optimized Poisson bounds replace the fixed-base bound when beneficial.

3. The original-law message error propagates through a whole ordered hierarchy with one factor $e^{t_* R}$, not one factor per level. Averaging its count dependence gives explicit finite all-count constants; the exponent budget uses the sharper range variance telescope.

These are source-specific analytic improvements. AM–GM, Poisson size bias, exponential tilting, Taylor remainders and Hoeffding's inequality are classical. Their names and general validity are not claimed as new Ram results.

The disadvantages are equally concrete. All-count constants in (28)–(29) may be prohibitive. A direct table expansion grows rapidly with count, grid size and depth. The L^1 premise itself must be certified under the original generator law, and arbitrary measurable kernels need not come with an effective approximation modulus. The improved small-exponent bound does not eliminate the source's structural transition or its tolerance ordering. Root-conditioned error bounds do not authorize replacing the root by fresh site data.

The source's finite rational equality remains a mathematical input to total computability. It uses selected conditional laws and subsequences, ultrametric representation, marked partition independence, endpoint passage, preserved closed records, finite stopping and final kernel collapse. Reading these arguments and checking selected local calculations is not an independent audit of every external theorem and its full proof. There is no assertion here that finite fixtures certify this entire chain.

Useful next work would implement the finite evaluator with directed rational arithmetic and keep an explicit certificate file listing the root law, rational table, exponents, cutoffs, retained interval and signed remainder. Sign-count-specific cutoff sets might reduce an A-tail enclosure further, but

would need exact retained probabilities. A posterior-weighted leaf error, using the exact identity (19), may beat the worst-case $e^{t_* R}$ factor when that density is itself rigorously bounded. None of these possibilities is a measured runtime claim or a computable rate for the source's compactness constructions.

9 Reading, Checks and Contribution

All 13 cached public text files were freshly displayed in bounded literal chunks, including every scientific section, the bibliography, figure code and README. The scientific hierarchy and equality arguments were read, not merely their headings. The figure's code and caption were read; its source rendering was not viewed. External papers cited inside the source are not thereby counted as read.

Fresh private original passages were the local-moment chapter lines 1–225, and TR250 lines 145–270 and 455–476. The first provides the actual Taylor remainder, normalization and polynomial-closure distinctions; the second provides finite functional coordinates and their limitations. Included subchapters were not opened. Core18 was only a catalogue locator and hashed endpoint, not a freshly read proof source and not relied on as an original derivation. Pre-read and post-read hashes, individual display intervals and all unread complements are recorded in the owned ledger.

Owned checks use rational finite packet factors and independent explicit rational enclosures for elementary functions where needed. Cases and assertions are separate units. They test the local formulas and bounds, not the threshold, the external ultrametricity theorem, or an entire hierarchy-completeness proof. No foreign checker was replayed. Public searches used only public titles and URLs; private text remained local.

Applicability stays L3. Contribution C3 is supported by the sign-sensitive A-range, asymmetric strict-margin ledger and depth-independent all-count law-preserving rationalization bound. No mathematical correction to the source is asserted in the scope examined. A valid conservative source bound is not an error merely because it admits refinement.

References

[1] OpenAI, *Computing the Random 3-SAT Threshold*, face date September 27, 2026. Assigned captured source, unchanged against the authenticated October 8 tree. In particular: `trials.tex`, `sensitivity.tex`, `transition.tex`, `closure.tex` and `certificates.tex`; all cached texts read.

[2] G. Carenini, *A polynomial scaling window for random k-SAT and a proof of the satisfiability conjecture*, ECCC TR26-229, October 5, 2026. Primary abstract and publication metadata checked October 9; complete proof not audited. https://eccc.weizmann.ac.il/report/2026/229/.

[3] J. Ding, A. Sly and N. Sun, *Proof of the satisfiability conjecture for large k*, Annals of Mathematics 196 (2022), 1–388. Publisher abstract and metadata checked, not the full proof. https://annals.math.princeton.edu/2022/196-1/p01.

[4] M. Subbarao, original confidential RamCalBooks, volume1-rev2, `ch03-local-moments-rev2.tex`, lines 1–225, especially the local differential representation, integral remainder and degree-closure proof. Included files were not read.

[5] M. Subbarao, original confidential `TR-rev-1-old-tr250.tex`, lines 145–270 and 455–476. Finite physical functionals, coordinate limitations and positive polynomial moments. The PDE mechanisms are comparisons, not SAT variational inputs.